

1. ICCM/DCCM DMA controller

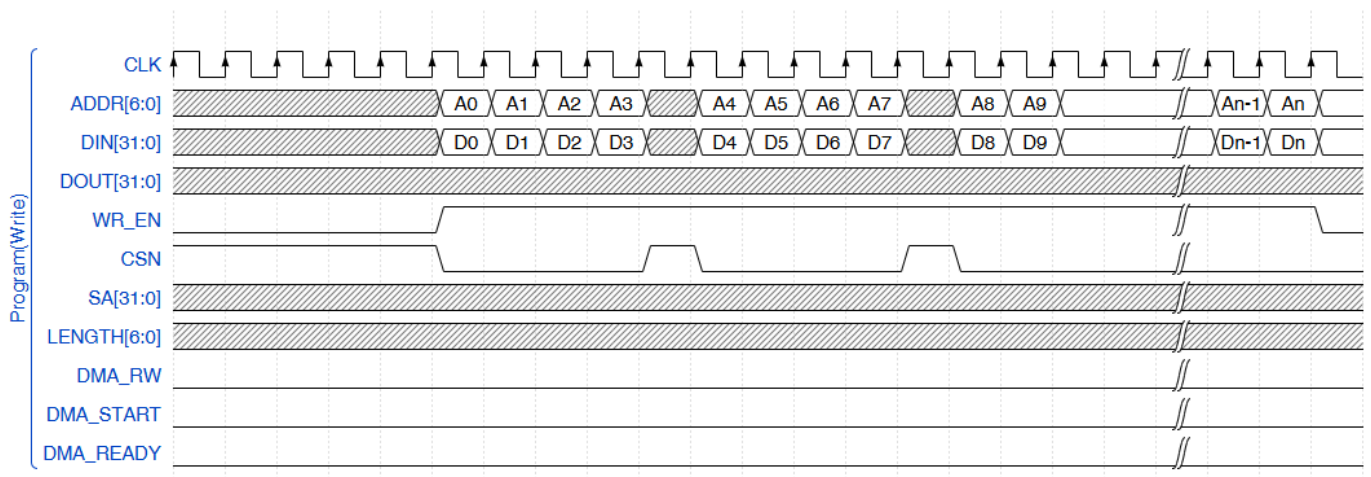
1.1 ICCM/DCCM DMA controller program I/F

NAME	Direction	Description
CLK	IN	Clock for OSPI, raising edge trigger
DMA_SA[31:0]	IN	DMA Starting address
LENGTH[6:0]	IN	DMA transfer data number = LENGTH+1(Unit: 32 bits)
DMA_RW	IN	0: DMA read, 1: DMA write
DMA_START	IN	1: Start the DMA transfer when DMA_READY = 0 0: Stop the DMA transfer, also clear DMA_READY
DMA_READY	OUT	DMA transfer is completed
ADDR[6:0]	IN	Address for DMA buffer
DIN[31:0]	IN	Data input for DMA buffer
DOUT[31:0]	OUT	Data output from DMA buffer
WR_EN	IN	0: Read cycle, 1: Write cycle
CSN	IN	Active low chip select for DMA buffer

1.2 DMA Write to ICCM/DCCM

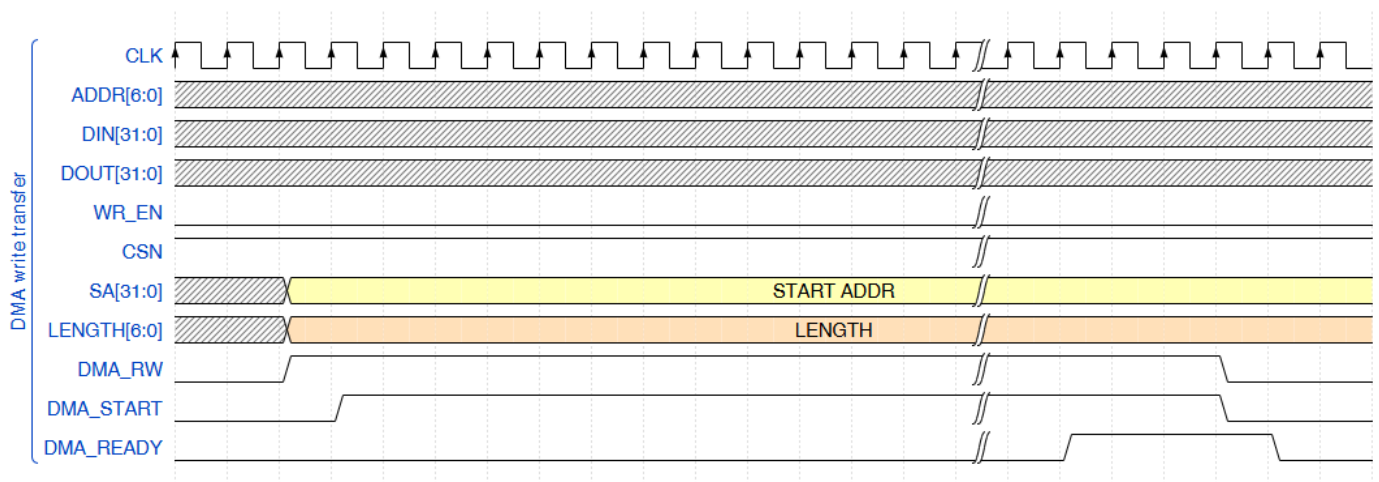
1. Step 1: Read data from DMA buffer

- Set ADDR[6:0] , WR_EN = H, CSN = L , DIN[31:0] for buffer writing.
- Fill all the data for DMA transfer.



2. Step 2: DMA write transfer to ICCM/DCCM

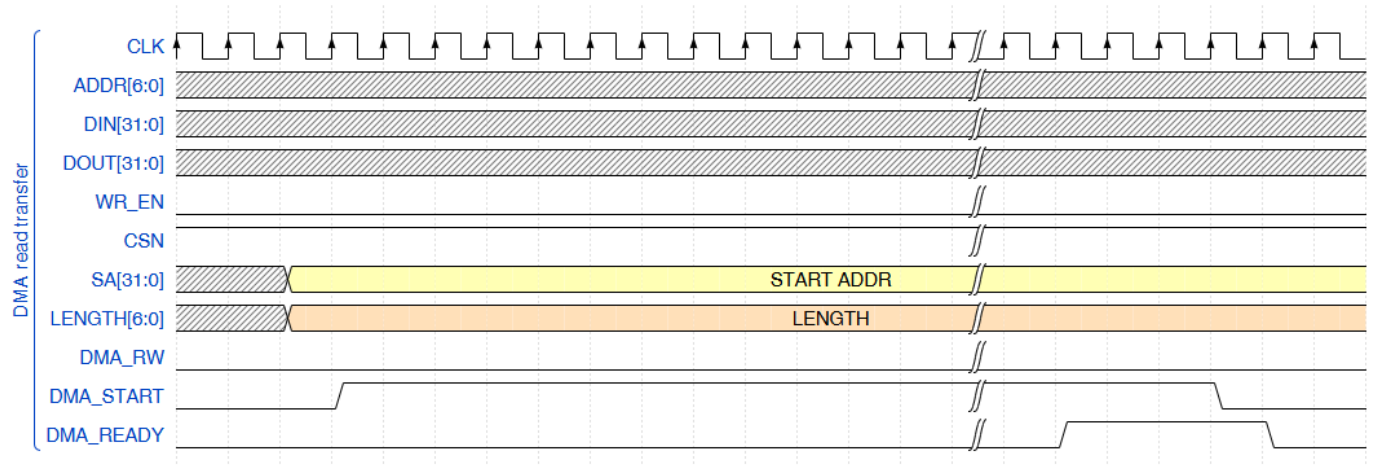
- Set Start Address/Length
- set DMA_RW = 1 for DMA write, set DMA_START = 0 to clear DMA_READY
- Assert DMA_START = 1 to starting DMA write transfer
- Waiting for DMA_READY = H(Done)



1.3 DMA Read from ICCM/DCCM

1. Step 1: DMA transfer from ICCM/DCCM

- V. Set Start Address/Length
- VI. set $DMA_RW = 0$ for DMA read, set $DMA_START = 0$ for clear DMA_READY
- VII. Assert DMA start to starting DMA read cycle
- VIII. Waiting for $DMA_READY = H$



3. Step2: Read data from DMA buffer

- I. Set $ADDR[6:0]$, $WR_EN = L$, $CSN = L$ for buffer reading.
- II. get DOUT at next cycle.
- III. Read all the ICCM/DCCM data by buffer read cycle.

